

RAMON ECUNG II

AI Engineer • Python & Applied AI Systems • 9+ Years Software Engineering • Active Secret Clearance

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SUMMARY

AI Engineer with 9+ years in software, systems integration, and automation across federal, local government, and enterprise environments. Hands-on experience with LLMs, AI agents, agentic workflows, embeddings, semantic search, MCP, and Docker. Builds reliable production systems with Python, SQL, PostgreSQL, HAproxy, Nginx reverse proxies, REST APIs, AWS Lambda, CI/CD, and containerized AI infrastructure.

TECHNICAL SKILLS

AI & Generative AI: Large Language Models (LLMs), local LLM inference, AI agent coding, agentic workflows, prompt engineering, context engineering, semantic search, vector similarity, retrieval architectures, multimodal AI, transcription, translation, image/video classification

AI Infrastructure & Tooling: LM Studio, GGUF models, MLX, Whisper/whisper.cpp, Nomic embeddings, vector embeddings, PostgreSQL/pgvector concepts, local model evaluation, inference pipelines, Dockerized AI services, Nvidia GPU and Apple Silicon inference

Python & Backend Engineering: Python, FastMCP, Flask, FastAPI, REST APIs, microservice/service-layer architecture, SQL, PostgreSQL, SQL Server, data processing, background jobs, integration engineering, automated testing.

Agents & AI-Assisted Engineering: Claude Code, Codex, coding agents, CLAUDE.md/AGENTS.md workflows, hooks, tool-oriented agents, implementation planning, autonomous task decomposition, long-running agent workflows, AI-assisted test generation and code review

Cloud & DevOps: AWS, Lambda, Docker Compose, Git, GitLab, GitHub, Jenkins, CI/CD, Linux, containerized services, production troubleshooting

Enterprise Automation: UiPath Studio, UiPath Orchestrator, REFramework, attended/unattended automation, queues, assets, and orchestration

Engineering Practices: Systems integration, API design, software architecture, unit/integration testing, code review, Git workflows, Agile/Scrum, Kanban, Jira, technical documentation, observability, incident response

SELECTED AI ENGINEERING PROJECTS

AI-Enabled Video Platform & Semantic Media System

Designed and developed an AI-enabled video platform architecture combining Python services, PostgreSQL, Docker, embedding-based search, automated media processing, and multiple machine-learning pipelines.

- Built semantic video search architecture using text embeddings/cosine similarity in vector space to retrieve media by conceptual meaning.
- Integrated embedding generation into the application data model and evaluated PostgreSQL/pgvector approaches for scalable search.
- Designed automated AI pipelines for transcription, translation, metadata generation, classification, tagging, and multimodal analysis.
- Developed architectures for facial recognition, identity matching, duplicate detection, partial matching, and content similarity analysis.
- Designed recommendation signals from viewing behavior, completion, abandonment, rewatches, and search to watch patterns.
- Architected async media processing workflows for HLS transcoding, thumbnails, AI inference, metadata extraction, and recovery.
- Designed service-layer APIs separating the web UI from media, search, recommendations, auth, and AI enabling apps and MCP usage.

Local LLM & Agentic Development Environment

Built and operated a local AI engineering environment for evaluating LLMs, coding agents, inference strategies, and autonomous software-development workflows.

- Deploy and evaluate LLMs on Apple Silicon systems with up to 64 GB unified memory using LM Studio, GGUF models, and inference tooling.
- Evaluate model size, quantization, context requirements, inference performance, and task suitability for software-engineering workloads.
- Design agent workflows that separate architecture/specification work from implementation, testing, review, and validation.
- Use structured agent instructions, repository-level context files, detailed implementation specifications, isolated worktrees, testing gates, phased commits, and validation checkpoints to improve reliability and consistency in autonomous coding workflows.
- Create workflows where larger reasoning models generate detailed engineering specifications suitable for smaller local models.
- Apply prompt and context engineering techniques to code generation, architecture analysis, and long-running implementation tasks.
- Evaluate local coding models across reasoning quality, context handling, tool use, quantization, and long-running implementation reliability.

AI Search, Embeddings & Retrieval Architecture

Semantic retrieval, vector search, hybrid indexing, and LLM tool integration

- Implemented and evaluated embedding-based semantic search using Nomic embedding models.
- Designed hybrid search architectures combining PostgreSQL full-text search, metadata filtering, and semantic vector retrieval.
- Designed search APIs supporting query execution, suggestions, facets, search history, and recommendation signals.
- Evaluated vector databases, SQLite with the Vec1 plugin, and pgvector against blob and relational PostgreSQL storage for embedding retrieval, balancing search performance, indexing, metadata filtering, operational complexity, and transactional consistency.
- Built FastMCP servers exposing application services as structured AI tools, enabling local LLM agents to interact with video and messaging platforms through MCP; implemented conversational semantic search, metadata retrieval, and messaging operations using local LLMs.
- Designed embedding lifecycle strategies for model changes, re-indexing, versioning, and maintaining consistent semantic search results across application updates.

PROFESSIONAL EXPERIENCE

OTOT – Senior RPA Consultant (embedded with UiPath/US Coast Guard FINCEN)

Oct 2024 – Present

Fully remote; Sole RPA developer for the United States Coast Guard Finance Center (FINCEN) on a federal engagement

- Own the complete software automation lifecycle as sole developer, including requirements analysis, architecture, implementation, testing, UAT, deployment, monitoring, and production support for mission-critical federal systems.
- Design and maintain production automation for 12 legacy-system processes, integrating enterprise applications and reducing manual processing by approximately 18 hours per week per automation.
- Build document-understanding and data-extraction workflows for ingestion, classification, extraction, validation, and downstream processing, eliminating approximately 24 hours of manual review each week across four operational units.
- Engineer reusable automation components, exception-handling strategies, logging standards, configuration patterns, and integration interfaces to improve reliability and maintainability.
- Design orchestration, queueing, retry, monitoring, and failure-recovery strategies for unattended production workloads.
- Integrate applications and services through APIs, structured data, databases, and enterprise systems while operating within federal security and deployment requirements.
- Work directly with stakeholders to translate operational problems into deployable technical systems and communicate production status, failures, and technical risks.

Sarasota County Government – Automation Developer (CJIS verified)

Dec 2023 – Oct 2024

Hybrid; UiPath architect who stood up the county's RPA practice

- Built the county's RPA Center of Excellence from the ground up as sole architect, establishing development standards, reusable components, intake processes, deployment practices, governance, and engineering controls.
- Designed automation systems end to end, translating business requirements into process designs, technical architecture, testing strategies, integrations, and production deployments.
- Introduced Git-based development, pull-request reviews, Kanban workflows, automated testing, reusable software components, and modern engineering practices to the automation program.
- Integrated enterprise applications, APIs, databases, and document/data-processing workflows into maintainable automated systems.

Cardinal Financial – Software Engineer II / Automation Developer

Jul 2019 – Aug 2023

Fully remote; Python automation and a cloud-native data warehouse

- Developed configurable Python and SQL automation systems for enterprise data workloads on AWS using Step Functions, ECS, Lambda, and PostgreSQL.
- Designed backend integration services and REST APIs using Python and Flask to connect enterprise systems and automate cross-platform workflows.
- Analyzed Python application and Pentaho ETL performance, identifying execution bottlenecks and code-complexity issues affecting enterprise data-processing workloads.
- Refactored automated tests and I/O-heavy processes to reduce redundant execution and improve test-suite performance.
- Built Python and Bash automation operating across macOS, Linux/Docker, and Windows environments.
- Created Docker-based development environments supporting both x86 and ARM64 architectures for consistent cross-platform execution.
- Built and maintained CI/CD pipelines with Jenkins and GitHub, including automated testing, version control, deployment validation, and production release controls.
- Produced system architecture, infrastructure, and process-flow documentation for AWS-based data platforms and formalized SDLC practices as the engineering team expanded.

UPC Insurance – Application Support/RPA Team Lead

Sep 2017 – Jul 2019

- Launched a new enterprise automation initiative, progressing from technology evaluation and infrastructure design through development and production deployment.
- Built and helped scale an RPA engineering team supporting automation development, requirements analysis, sprint planning, and production operations.
- Designed and developed C# integration services transferring financial data between SaaS and self-hosted enterprise systems.
- Built automated document-processing workflows using C#, OCR, regex extraction, and classification rules to extract claims data and identify high-priority correspondence.
- Built operational time-series monitoring dashboards and automated reporting to improve visibility into the production system.
- Wrote PowerShell and VB.NET monitoring tools that detected production and UAT system failures and automatically generated operational alerts and support tickets.
- Worked across applications, infrastructure, APIs, vendors, and internal engineering teams to troubleshoot production issues and minimize service disruption.

CERTIFICATIONS & EDUCATION

UiPath Certified Advanced RPA Developer (2019) • UiPath Certified RPA Associate (2024) • Python Institute: PCAP (2022), PCED (2024)

B.S., Information Sciences and Technology – The Pennsylvania State University • Dean's List, 3.92 GPA